

Garmin GPS 18 OEM. It is WAAS-enabled and available in CMOS-level serial or USB 2.0 full-speed versions. It comes with an integrated magnetic base, which is sufficient to provide an extremely high level of accuracy.

Motor control. The UGV employs Robot-Solutions' RS160D motor control. RS160D servo uses custom software to implement a 2.5-channel high-powered servo system. The servo uses a quadrature encoder feedback mechanism to control torque, velocity and position of the output. It operates in pulse-width-modulation (PWM) mode communicating serially with the processor through an RS232 cable.

Data is sent in ASCII format from the processor using Visual C++ platform. RS160D is a two-channel controller. Each channel controls two different motors (right and left).

Data is sent as:

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"@0ami" selects mode 1, i.e., PWM mode,
"@0s30" selects serial communication
mode
"@0st255" selects torque control
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Differential drive is used to turn the vehicle left or right depending on the data received from the sensors. Torque values for the controller vary

between -255 and +255. Different speeds were tested for the two motors in order to optimise the degree of curve to which the vehicle has to turn in order to avoid obstacles and stay within the lane. The table shows a sample of values that were tested on the motors and their upshots.

Sample Values Tested on the Motors and Their Upshots

Left motor	Right motor	Uphshot
@0ST120	@0ST120	Both motors off
@0ST130	@0ST130	Low voltage, vehicle moves with jitter motion, not stable
@0ST170	@0ST170	Vehicle moves straight smoothly at average speed of 2 Mph
@0ST170	@0ST140	Vehicle moves right at an angle
@0ST140	@0ST170	Vehicle moves left at an angle

Mechanical design overview

The split-frame design ensures better locomotion capabilities of the vehicle. The front wheels are independent and can rotate 360 degrees. This enables the vehicle to turn sharply when it gets very close to any obstacle.

The design also permits the UGV to negotiate ramps in a better way. The split-frame technology enables independent motion of the two frames and hence the ramp climbing becomes easier, reducing the burden on the motors. Even if one of the four wheels gets stuck in a pothole, the technology ensures that the other three wheels are placed on the ground. This helps

the UGV to negotiate uneven surfaces and hence the use of the encoder is evaded.

The design is done on SolidWorks. The UGV is fabricated using mild steel material and is designed to take 150kg load. The material is laser-cut and welded together and then the painting is carried out (refer Fig. 2).

Electrical design

Block diagram of the UGV's electrical system design is shown in Fig. 3. All the components have a common ground. For safety, emergency stops (E-stops) are connected in series with the circuit. These disconnect the power supply to the circuit with a single push of a button.

The central custom-made power distribution board is made of wood as it has electrical insulating property. Two metallic terminals are firmly mounted on the board and connected to a 12V, 34Ah power source. This connection is made using 40A, thick copper wires. Solar cells are used to recharge the battery.

The other connections from the motors and supply board to the motor controller, mechanical emergency-stop (E-stop) and wireless E-stop are made using 14AWG wires.

Pressing the mechanical E-stop switch turns off the supply to all the components. A normally-closed (N/C) switch by default, the mechanical E-stop is placed at the back of the robot at an accessible position.

The wireless E-stop switch has a range of about 2.5 metres (100 feet). It employs an inbuilt relay to switch contacts between normally-closed (N/C)

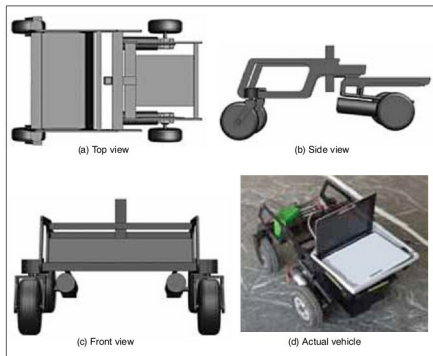


Fig. 2: The UGV fabricated using split-frame technology